Satyen Patel

Email: satyen.patel@students.iiserpune.ac.in
Personal email: satyen2003@gmail.com
Phone: +91 7203822656
Languages: English, Hindi, Gujarati(Native)

Research Interests: Poisson geometry, symplectic geometry, deformation quantization, sigma models and BV/AKSZ formalism, Courant algebroids, homological and categorical methods in mathematical physics.

Education

BS–MS Dual Degree in Mathematics *August 2021 – May 2026 (Expected)*
Indian Institute of Science Education and Research, Pune
CGPA: 8.1/10
MS Thesis Evaluation: 98/100

Relevant Coursework: Differential geometry and Lie groups, Algebraic topology, Algebraic geometry, Functional analysis, Rings and Modules, Quantum Field Theory, Gravitation.

Independent Study and Reading Projects: Poisson geometry, coisotropic and Dirac reduction, Lie algebroids and groupoids, Poisson Sigma model, BFV/BRST reduction.

Research

Master's Thesis at International Centre for Theoretical Sciences (ICTS-TIFR) *April 2025 – present*

Advisor: Dr. Rukmini Dey (rukmini@icts.res.in)

Title: *Geometric and Deformation Quantization through Examples*

- Covers geometric quantization of coadjoint orbits and Kähler manifolds, the Fedosov construction on symplectic manifolds, and the formality machinery of Kontsevich. Includes an original result extending Fedosov-type quantization to Poisson manifolds with quantum moment maps.

Academic Visitor (upcoming), International Centre for Theoretical Sciences *May 2026 – Aug 2026*

Collaborator: Dr. Rukmini Dey

- Constructing explicit Fedosov-type star products on Lagrangian embeddings in $\mathbb{C}P^n$, with the aim of connecting deformation and geometric quantization on Poisson manifolds via Berezin-type methods and pullback reproducing kernel Hilbert spaces.

Semester Project on Topics in QFT *Aug 2023 – July 2024*

Advisor: Dr. Sachin Jain, IISER Pune

- Studied supersymmetric quantum mechanics and the super-Poincaré algebra via matrix models. Applied Landau equations to analyze singularities of multi-loop integrals.

Summer Reading Project in Geometry *May 2023 – July 2023*

Mentor: Dr. Mainak Poddar, IISER Pune

- Studied differential geometry with applications to general relativity.

Preprints

- S. Patel, *Fedosov quantization of Poisson manifolds and quantum moment maps*, under review at *Letters in Mathematical Physics* (submitted March 2026). Extends Fedosov-type deformation quantization to Poisson manifolds with group actions, using the formal symplectic groupoid.

Fellowships

- KVPY Scholar 2021–2026
- Long Term Visiting Student Program (LTVSP), ICTS-TIFR, 2025–2026

Talks

- *Star products on Poisson manifolds from the Poisson Sigma Model*
  ICTS In-House Symposium 2026, 6–8 April 2026
- *Fedosov quantization on smooth manifolds and hidden symmetries*
  ISQGD Special Session: Analysis on Manifolds, Online, 14 March 2026.

Programs and Workshops

- *Selected participant*, From Hochschild homology to topological Hochschild homology
  International Centre for Theoretical Sciences (ICTS-TIFR), *upcoming*, 25 May – 5 June 2026